

COVID-19 SPRI Assay on Freedom EVO100

Automated Liquid Handling Configuration:

Freedom EVO100

Robotic Arm Left (RoMa Standard)

Multi-channel Arm 96 (MCA96)

Inheco Heat & Cool Modules (CPAC) with 96 PCR Adapters – 2 units

Heat & Cool Shaker (Thermoshake) with 96 PCR Adapter

Carriers and Racks:

MCA96 Tips Adapter 4 positions – 2 units

MP 4 positions flat – 1 unit

Invitrogen Strip Magnetic Plate – 1 unit

Consumables (for 96 samples):

MCA96 SBS 150µl Filter Tips – 3 boxes

MCA96 SBS 50µl Filter Tips – 3 boxes

96 DWP – 1 unit

96 Round Plate – 2 units

Full Skirted 96 PCR Plate (BioRad / 4Titude) – 2 units

MCA96 Trough 250ml – 1 unit

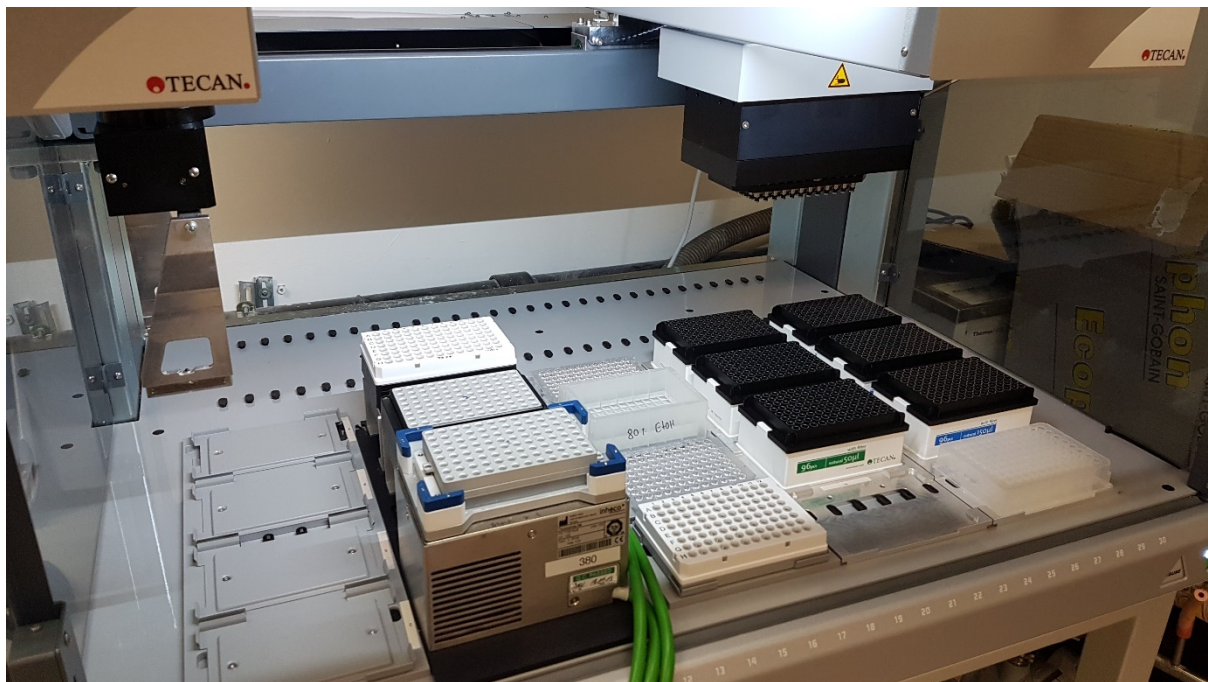
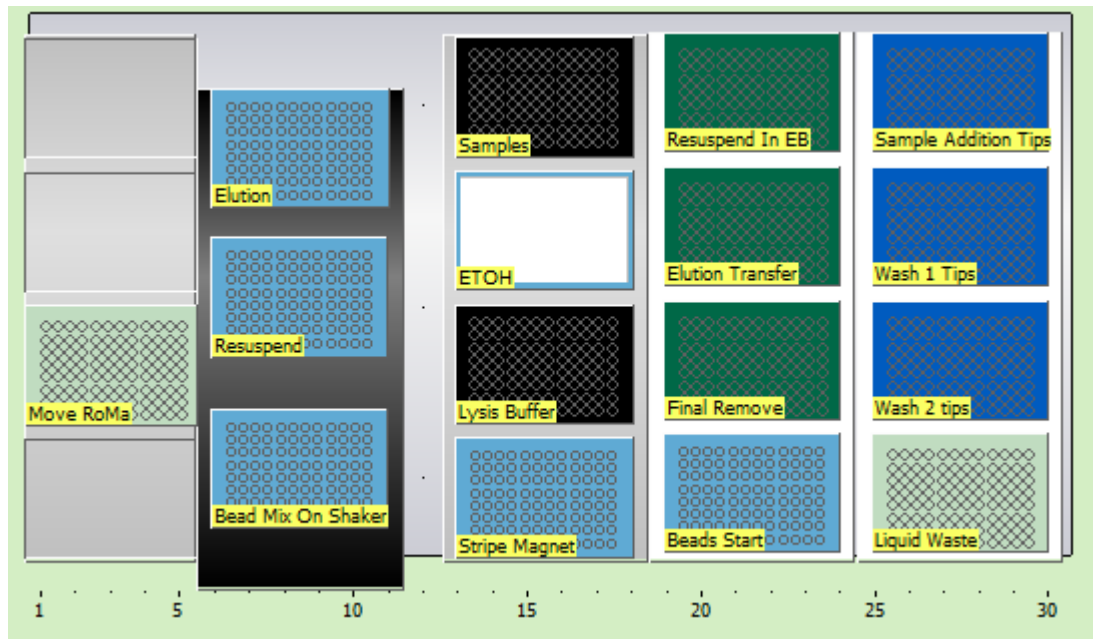
Solutions:

SPRI Beads

Lysis/binding buffer

ETOH 80%

Elution Buffer (10 mM TRIS pH 8)

Working table Layout:

Assay Process:**Pre-Actions:**

1. Heat the Thermoshake to 25°C
2. Cool CPAC Rear to 4°C
3. Cool CPAC Middle to 10°C
4. Load 3 MCA96 150µl Filter Tips, 3 MCA96 50µl Filter Tips, Empty 96 DWP, 96 Round plate with 50µl Samples, 96 Round plate with 34µl Lysis Buffer, MCA96 Trough with 40ml 80% ETOH, 96 PCR plate with 45µl Beads and 96 PCR with 22µl EB Buffer (according to the working table)

Sample Addition & Beads Mix:

1. Transfer the 96 PCR Beads plate from the working table to the Thermoshake
2. Transfer 28µl lysis buffer to the Beads plate.
3. Transfer 28µl Samples to the Beads and Lysis plate.
4. Start Timer (10 minutes)
5. Mix the Beads & Lysis & Sample (80µl Mix)
6. 4 cycles of Mix with 2 Min interval (80µl Mix)
7. Wait 10 minutes from step 4

Magnet Step:

1. Transfer the plate to the Invitrogen magnet
2. Set the Thermoshake to 30°C
3. Wait for 2 Min on the magnet
4. Aspirate the supernatant to the DWP waste (110µl) (use 150µl Tips – Box 1)

ETOH Wash:

1. Add 110µl of ETOH to the plate (on the magnet) (use 150µl Tips – Box 2) (do not return the tips)
2. Use the robotic arm and move the plate on the magnet (center – left – center – right – center total of 5 cycles). This step washes the beads by moving them from the left to the right side of the well through the liquid
3. Aspirate the supernatant to the DWP waste (110µl) and drop the tips
4. Perform steps 1-3 two times
5. Final ETOH aspiration: aspirate 20µl of leftover to the waste (use 50µl Filter Tips – Box 1)

Beads Air Dry & Elution:

1. Transfer the plate to the Thermoshake (30°C) for 2 Min.
2. Transfer the plate to the Middle CPAC (10°C)
3. Set the Thermoshake to 10°C
4. Add 20µl of EB Buffer to the PCR plate (on CPAC Middle) (use 50µl Filter Tips – Box 2, do not return the tips)

5. Start Beads Resuspension:
8 Cycles of:
Aspirate 15µl from the center and dispense on the right side
Mix (aspirate and dispense) 17µl – 10 cycles
Aspirate 15µl from the center and dispense on the left side
Move the plate to the Thermoshake and shake for 30Sec linear shake 1,000rpm
Return the plate to the CPAC
6. Drop the tips
7. Make sure the beads are fully resuspended
8. Transfer the plate to the Invitrogen magnet
9. Wait for 60 Sec
10. Transfer 18µl from the plate (on the magnet) to a new 96 PCR plate (on the rear CPAC, 4°C) (use 50µl Filter Tips – Box 3)

RNA Plate is ready for downstream analysis

When Done turn off the Thermoshake and two CPAC units